

Shadow the most appropriate answer(s) for each of the following questions into the above bubble sheet:

- 1- The following data platform is an example of an SQL database that is typically used for applications:
a- Web Tier
b- Schema
c- PostgreSQL ✓
d- SQL
- 2- The key concept behind distributed applications systems is that they are software applications to only ensure there is a APIs tier for system integration:
a- True
b- False
- 3- Two of the very key motivations to build distributed systems are (select two):
a- Scalability to cater for workload growth
b- Scalability to reduce down time
c- Availability to cater for workload growth
d- Availability to reduce down time
- 4- TCP/IP sockets are used as the de facto standard communication interface for distributed applications runs beneath different technologies like JDBC, JMS, and REST APIs.
a- True
b- False
- 5- Distributed applications can be built in different topologies with key features to enable an application to run over different physical or virtualized nodes. Which of following are examples of such architecture (select all that apply):
a- Microservices
b- Master-Slave Systems
c- Multi-Tier Web Applications
d- Peer-to-Peer Systems
- 6- The 12-Factor Apps is a framework used for socket programming and network interfacing.
a- True
b- False
- 7- One of the main concepts behind using public cloud model is to pay as you utilize the services instead of buying and owning the infrastructure components needed for those services.
a- True
b- False
- 8- The most important tier in Multi-Tier applications where the patient, hospitals, diagnostics, and other data and records are physically stored is:
a- EMR
b- Database
c- EPR
d- AWS
- 9- Amazon Web Services (AWS) is a collection of components including infrastructure, applications, databases, web servers, among others offered in highly available, manageable, and integrated services.
a- True
b- False
- 10- When architecting distributed applications, two types of requirements are applied, Functional and Non-Functional requirements. Which of the following is essential one for the Functional Requirements?
a- Server Logs
b- Reliability
c- UI/UX
d- Availability

11- The core protocol for enabling modern distributed applications to be designed and implemented including different patterns like client/server, n-tier and p2p is:

- a- UDP
- b- ADSL
- c- TCP/IP
- d- ATM

12- One of the key non-functional requirements in distributed system that enables surviving failures:

- a- Concurrency
- b- Scalability
- c- Fault Tolerance
- d- APIs

13- ATM machines are an example of master tier in the Master-Slave architectural model.

- a- True
- b- False

14- Distributed applications architecture includes four layers. The following are three of them (select three):

- a- Presentation
- b- Transactions
- c- Application Processing
- d- Database

15- In distributed component architecture, the component that lays the central role in enabling other components to communicate together:

- a- Web Server
- b- Middleware
- c- FAT Client
- d- Database

16- The architecture of EHR doesn't require a database tier as it just connects to other medical applications.

- a- True
- b- False

17- BitTorrent and Bitcoin network are examples of the following distributed applications architecture:

- a- N-Tier Client-Server
- b- Peer-to-Peer
- c- Client-Server
- d- Master-Slave

18- The TCP/IP protocol contains several layers used for completing connectivity. While TCP is connectionless, UDP is connection-oriented.

- a- True
- b- False

19- In distributed applications, two models of communication can be utilized which are Synchronous and Asynchronous. Which of the following are attributes of the Asynchronous model (select all that apply)?

- a- Sender and receiver can use different sockets on different nodes
- b- Sender and receiver don't have to be connected at the same time
- c- Sender and receiver must use the same programming and database technologies
- d- Sender and receiver must be connected at the same time

20- Electronic Health Records (EHR) applications' main role is to provide replacement for other medical applications like HIMS, PACS.

- a- True
- b- False

- 21- "Presenting each user of the software with the impression that they are working with their own copy of the system while, at the same time, making efficient use of system resources" is called:
- a- Availability
 - b- Multi-tenancy
 - c- Service Configuration
 - d- Complexity
- 22- The Local Zones feature in AWS is used for global content delivery.
- a- True
 - b- False
- 23- In Socket programming, the socket is defined by the following elements on both communicating sides:
- a- IP Address and MAC number
 - b- IP Address and Port number
 - c- IP Address and GUID number
 - d- IP Address and Socket number
- 24- One of the main methods of interactions among distributed systems components is procedural based interaction. The following is an example of such type:
- a- RPC
 - b- JMS
 - c- MQ Systems
 - d- Web Tier
- 25- When working with a system like Zoom for online meetings and conferencing, the only way to access the application is through a thin web-based client.
- a- True
 - b- False
- 26- The IaaS model provides virtual machines and storage but requires administrators to manage the OS.
- a- True
 - b- False
- 27- The purpose of environment variables in backend deployment is to securely manage API keys.
- a- True
 - b- False
- 28- Which component is responsible for business logic in a Telemedicine Application:
- a- Frontend (Next.js)
 - b- Backend (Node.js)
 - c- Database (PostgreSQL)
 - d- Vercel hosting
- 29- The primary advantage of using the PaaS cloud computing model compared to other models is:
- a- Full control over infrastructure
 - b- No need to manage servers or runtime environments
 - c- Ready-to-use software applications
 - d- Built-in patient monitoring tools True
- 30- The following transport layer protocol guarantees ordered and error-checked data delivery:
- a- UDP
 - b- TCP
 - c- HTTP
 - d- FTP